Battambang City, Battambang Province

Heat is the killer here, and its worst weeks, March to May, fall partly after the forecast ends. It finds older people at home and farmers working off debt in the midday sun. List who needs a daily check now, and give households time on their loans before the dry season bites.

REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
Batdâmbâng (Battambang) province, province level	400 forecast, 320 of them January–March; 150 already past (Jun–Sep). The highest province in Cambodia	6-month forecast, Oct 2026 – Mar 2027; April and May, usually hotter, are not covered	Heat in March–May; water shortage in peri-urban communes; crop loss and debt; dengue when the rains return

16/16 recommendations checked. All 10 actions were corrected, none unverified; 6 contraindicated (the six "Do not do this" cards). All 9 For you steps were checked: 4 confirmed and 5 corrected. Two lens proposals were dropped in verification: ceramic water filters (the city's new piped network reports 94% coverage, and no stockist was found) and paying for dengue control from the heat budget (dengue peaks in the rains). A claimed 2026 collapse in the Tonle Sap fish catch turned out to be a 2020 article. Exposure figure: Climate Impact Lab El Niño Heat Mortality Tracker data, version of 26 October 2026, pre-peer-review, methods forthcoming. It is published for the whole of Battambang province,

EXECUTIVE SUMMARY — AT A GLANCE



13.10°N 103.20°E, Cambodia · Natural Earth

- **The risk.** Battambang is the province with the most forecast heat deaths in Cambodia; April 2024 reached about 43°C nationally, yet Phnom Penh recorded zero heat deaths, because they are not counted.
- **For you.** Now: if your income fell after returning from Thailand or losing a crop, ask your commune for a free IDPoor assessment, so any cash or fee waiver can find you.
- **The money.** KHR 102.0M (US$25,000) to KHR 550.9M (US$135,000); daily checks on older people at risk buy the most lives.
- **Do not.** Do not take a new loan to replant a lost crop: with loan relief ended, a failed dry season turns debt into a lost land title.

For you, as a resident	9 steps	p. 2
What to do	**10** actions	p. 6
Who does what	RACI chart	p. 13
Where money does the most	funding	p. 13
Do not do this	**6** items	p. 15
What has happened here	the record	p. 17
How this was produced	methodology	p. 19
References	sources	p. 21

* Suggested formulation: check with a health worker before sharing.

not for the city, and it counts heat only. The city's share was not calculated. The forecast stops in March, and Battambang's hottest months are April and May, so the toll for this season is likely higher than the figure shown. No pre-arranged money was found for Battambang: no Red Cross early action protocol, no SEADRIF policy, and no published heat or drought trigger. The funding target fell by more than a third in verification, from KHR 612.1 million (US$150,000) to KHR 387.7 million (US$95,000), because two placements duplicated the new water network or fell outside the heat season.

FOR YOU, AS A RESIDENT

9 STEPS

For a family in Battambang city or a nearby rice commune, El Niño arrives as a long, hot dry season. The danger peaks from March to May, when it is hottest, ponds run low in the communes, and loan repayments fall due after a poor harvest. Older people at home, people with high blood pressure or diabetes, and anyone working in the fields at midday are most at risk. The ranked actions below are written for the province, the city, and their partners. These are the ones you can take yourself.

If your income fell, after returning from Thailand or losing a crop, ask your commune or sangkat for an IDPoor assessment

PARTIALLY CONFIRMED | FROM ACTION 4

BY WHEN	COST TO YOU	PROTECTS
now; aim for 31 December 2026	none	access to cash and free health care

On-Demand IDPoor has been national policy since 2020 and is free. An Equity Card waives fees at public health centres and hospitals, and the Family Package pays IDPoor households with a pregnant woman, a child under two, a person over 60 without a pension, or a person with a severe disability. Any emergency cash in a drought would also go through IDPoor. There is no deadline; 31 December is a target so you are on the list before the hot season.

If you have a loan you may not be able to repay, ask your lender in writing to restructure it before you miss a payment, and do not borrow again to replant

PARTIALLY CONFIRMED | FROM ACTION 3

* Suggested formulation: check with a health worker before sharing.

BY WHEN	COST TO YOU	PROTECTS
before the first missed payment; by 15 February 2027 at the latest	none	your land title

The central bank's blanket loan relief ended in 2025 and was not extended, but lenders can still restructure loans case by case. In March 2026, 71% of returning migrant workers surveyed were in debt, averaging about KHR 22.4 million (US$5,500). If the lender refuses, use its complaint channel through the Cambodia Microfinance Association or the Association of Banks in Cambodia, or the central bank's consumer protection line; LICADHO and CENTRAL can refer you.

Store drinking water first, before March, in clean covered containers, and boil or treat it *

CONFIRMED FROM ACTION 5

BY WHEN	COST TO YOU	PROTECTS
1 March 2027	your own containers	the household

In 2016 water shortages peaked in May, and poor households paid the highest prices. On 19 October 2026 the water ministry ordered that household water take priority over dry-season rice. In the city, the new piped network reports 94% coverage, but households in the rice communes outside it should store water early.

Put guppy fish in every large water jar and keep lids on smaller ones

PARTIALLY CONFIRMED FROM ACTION 9

BY WHEN	COST TO YOU	PROTECTS
31 January 2027, and keep it up through the rains	small; ask your health centre or village health volunteer where to get fish	children, from dengue

Household water jars are where dengue mosquitoes breed. In a Cambodian trial, guppies plus community engagement roughly halved adult dengue mosquitoes (Hustedt 2021), and in an earlier study jars with guppies had 79% less infestation (Seng 2008). Cambodia has had two big dengue years in a row: 63,106 cases in 2025, and 2026 up 58% by July.

* Suggested formulation: check with a health worker before sharing.

Register free with 1294, Cambodia's early warning system

PARTIALLY CONFIRMED · ROLE-SPECIFIC

BY WHEN	COST TO YOU	PROTECTS
now	none	the household

The 1294 system, built by People in Need and run by the provincial disaster committees under NCDM, sends voice warnings to registered phones in every province. It is mainly a flood system; heat and drought alerts are sent only where issued, so also follow the water ministry's temperature forecasts on television and Facebook.

Take older relatives for a blood pressure check and a medicine review at a public health centre in February*

PARTIALLY CONFIRMED · FROM ACTION 7

BY WHEN	COST TO YOU	PROTECTS
February 2027	free with an Equity Card; otherwise ask the health centre, fees are usually small	older people

About one adult in five in Cambodia has raised blood pressure and about one in sixteen has diabetes (STEPS 2023). Heat kills people with these conditions first, and some medicines, such as water tablets, need adjusting in hot weather. Ask the health worker about each one.*

Agree with a neighbour who will check each day, from March to May, on anyone older, bedridden, or living alone

CONFIRMED · FROM ACTION 1

BY WHEN	COST TO YOU	PROTECTS
agree by 28 February 2027	none	older people and the bedridden

In studies of heat-wave deaths, being confined to bed raised the risk of dying more than sixfold, and not leaving the house daily more than threefold (Bouchama 2007). The health ministry has advised checking people over 65 at least twice a day in hot weather.

* Suggested formulation: check with a health worker before sharing.

On the hottest days, use a fan with wet skin, rest at midday, work outdoors early, and know the signs of heat stroke

CONFIRMED · ROLE-SPECIFIC

BY WHEN	COST TO YOU	PROTECTS
March – May 2027	none	everyone, and outdoor workers most

A fan with damp skin helps in Battambang's heat, which usually peaks at 35–37°C in April; above about 38°C, move older people to a cooler place. Heat stroke shows as confusion, hot dry skin, or collapse: cool the person with water and call 119 for an ambulance, and keep the provincial referral hospital's own number, as ambulances outside Phnom Penh can be slow.*

When the first storms come in April and May, get off open paddies and out of water at the first thunder

CONFIRMED · FROM ACTION 9

BY WHEN	COST TO YOU	PROTECTS
April – May 2027	none	farmers and children

Lightning killed 68 people in Cambodia in 2024, 84 in 2023, and 63 in 2022, many of them in open fields at the start of the rains. The government linked more strikes in 2024 to the El Niño heat.

What to ask of the people in charge

- Ask your village chief or health volunteer whether your village's list of older and bedridden people is made, and who will check on them from March (action 1).
- Ask the Provincial Department of Education whether the hot-season school schedule is signed before March, so it starts without delay (action 2).
- Ask your lender, through the Cambodia Microfinance Association or the Association of Banks, for a March–May repayment pause for drought- and flood-hit borrowers (action 3).
- Ask the provincial disaster committee (PCDM) which day water trucks will reach your commune's health centre and school if ponds run dry (action 5).

* Suggested formulation: check with a health worker before sharing.

WHAT THIS PLACE SHOULD DO RANKED

Ranked by expected lives saved per unit of effort before the peak. Backing lenses are shown per action; an action several lenses reached independently is the stronger bet.

1. List every older person who is bedridden, housebound, or alone by 31 January, and run daily checks from March to May whenever the forecast reaches 40°C

PARTIALLY CONFIRMED | PUBLIC HEALTH · FRONTLINE · PHILANTHROPIST · EMERGENCY MANAGER | ADAPTATION INVENTORY: NOT CHECKED

OWNER	BY WHEN	COST	PREVENTS
Provincial Health Department, through village chiefs, village health support group (VHSG) volunteers, and HelpAge Cambodia's older people's associations where they are active	list by 31 January 2027; checks March – May 2027	KHR 102.0 million (US$25,000) pilot for about 5,000 people; KHR 224.5 million (US$55,000) if city coverage is shown (placements 1–2)	deaths of older people alone at home in the heat

In six case-control studies of heat-wave deaths, being confined to bed carried an odds ratio of 6.44 and not leaving home daily 3.35, while visiting cool places was protective (0.34) (Bouchama 2007). Ahmedabad's heat action plan, which triggered at a forecast above 40°C, averted an estimated 1,190 deaths a year (95% CI 162–2,218) (Hess 2018). Cambodia has no colour-coded heat warning, so the trigger has to be a set temperature on the water ministry's forecast. Volunteers carry ORS, refer people to health centres, and can take them to a cooler place, such as a health centre or pagoda, on the worst days.

Every village has one or two VHSG volunteers. HelpAge reports 156 older people's associations with 7,800 members across Battambang, Banteay Meanchey, and Siem Reap, mostly rural; its coverage of the city's sangkats is not documented. Grandparents are the main carers in about two thirds of migrant households, and the 2025 border closure brought many working-age adults home, often in debt.

* Suggested formulation: check with a health worker before sharing.

Transferability: the evidence comes from France, the United States, and India, not Cambodia, and the Ahmedabad result is for a whole-city plan, not home visits alone. A Battambang survey found migration had only modest effects on older people's care, so the list should be built door to door, not assumed.

2. Sign the hot-season school schedule (7–10am and 2–5pm) before 1 March, so it starts automatically when the forecast passes 40°C

PARTIALLY CONFIRMED | EMERGENCY MANAGER · HISTORIAN | ADAPTATION INVENTORY: NOT CHECKED

OWNER	BY WHEN	COST	PREVENTS
Provincial Department of Education, Youth and Sport, under the ministry (MoEYS)	signed by 28 February 2027	none	children in class through the hottest hours while a decision is awaited

In late April 2024 the ministry ordered classes from 7 to 10am and 2 to 5pm, with breaks and drinking water, citing temperatures above 40°C between 10am and 2pm. It was a one-off notice issued weeks into the heat, with precedents in 2016 and 2019, and normal hours resumed with the rains. A pre-signed order with a temperature trigger would be new and needs a provincial or ministry decision.

Transferability: local precedent, not a study of deaths. It costs nothing and removes a delay.

3. Ask the central bank and lenders for a March–May repayment pause for drought- and flood-hit borrowers, modelled on the 2025 relief for returning migrants

PARTIALLY CONFIRMED | ECONOMIST · SKEPTIC · FRONTLINE · FOOD SUPPLY CHAIN | ADAPTATION INVENTORY: NOT CHECKED

OWNER	BY WHEN	COST	PREVENTS
National Bank of Cambodia, with the Cambodia Microfinance Association and the Association of Banks in Cambodia	ask by 31 December 2026; pause March – May 2027	no public cost; lenders carry the deferred interest	distress land sales and farmers working through the heat to make repayments

* Suggested formulation: check with a health worker before sharing.

The central bank's forbearance ended at the end of 2025, and on about 1 September 2026 it refused lenders' request to extend it. Bank gross non-performing loans were 9.6% in June 2026 (Moody's, via AMRO). A model exists: lenders pledged restructuring for returning migrants in June 2025, the central bank issued a relief directive on 10 December 2025, and microfinance institutions extended it to 31 March 2026. Remittances from Thailand fell by about 27% in 2025 (central bank), and 85% of indebted returnees surveyed could not repay (CENTRAL, March 2026).

Most microloans are secured on land titles. Asking for time costs the public nothing and is cheaper than any cash transfer.

Transferability: a domestic precedent exists, but a pause for drought and heat would be a new ask, not an existing scheme. No study links repayment pressure to heat deaths; the link is that people work through the heat to repay.

4. Run IDPoor catch-up assessments in every commune for households whose income fell, and agree a drought trigger in PRISM that would release a cash top-up

PARTIALLY CONFIRMED | ECONOMIST · FRONTLINE · FOOD SUPPLY CHAIN · PHILANTHROPIST | ADAPTATION INVENTORY: NOT CHECKED

OWNER	BY WHEN	COST	PREVENTS
Ministry of Planning through commune and sangkat councils; NCDM with WFP for the trigger; MoSVY and the social protection fund for payment	**assessments by 31 January 2027; trigger agreed by 28 February 2027**	**assessments within existing budgets; top-up unfunded**	**returnee and flood-hit families missing every payment because they are not on the list**

On-Demand IDPoor lets households that have become poor since the last survey be assessed at the commune, and the IDPoor portal launched in March 2026. In 2022–23, KHR 222 billion (US$55.7 million) went to 495,307 "at-risk" households across an inflation transfer and a flood transfer (UNDP); the flood transfer alone reached 99,169 households. PRISM, run by NCDM with WFP, uses IDPoor data, but its documented use is flood mapping; no heat or drought cash trigger has been published.

* Suggested formulation: check with a health worker before sharing.

Transferability: the delivery channel is proven for floods; for drought or heat it is an advocacy ask with no payer yet. Rice markets are working and supply is ample, so cash rather than food is the right form.

5. List the peri-urban communes whose ponds will run dry by May, and pre-contract water trucks for their health centres, schools, and poorest households by 31 January

PARTIALLY CONFIRMED | EMERGENCY MANAGER · HISTORIAN | ADAPTATION INVENTORY: NOT CHECKED

OWNER	BY WHEN	COST	PREVENTS
Provincial Committee for Disaster Management (PCDM), with the Provincial Department of Water Resources and Meteorology (PDWRAM)	list by 15 December 2026; contracts by 31 January 2027	KHR 61.2 million (US$15,000) contingency (placement 5; estimate)	health centres without water and poor households paying gouged prices

In 2016, 37% of households surveyed nationally reported water shortages by May, and the response came after shortages were declared (WFP). In July 2026 NCDM ordered water trucks, pumps, and tanks for schools, hospitals, and health centres, and on 19 October the water ministry put household water ahead of dry-season rice. The city itself is now largely on piped water: the new treatment plant opened on 30 April 2026 and the network reports 94% coverage of about 219,810 households. The need is in the rice communes outside it. Confirm the waterworks' April–May intake margin too, since low dry-season intake was flagged before.

* Suggested formulation: check with a health worker before sharing.

Transferability: local precedent from 2016. The cost is an estimate, and no record of the Red Cross branch trucking water in Battambang was found.

6. Publish reservoir volumes and commune limits on dry-season rice for Sangkae and Ek Phnom by 15 November, and pay no fuel or pumps for second or third crops

PARTIALLY CONFIRMED | FOOD SUPPLY CHAIN · SKEPTIC | ADAPTATION INVENTORY: NOT CHECKED

OWNER	BY WHEN	COST	PREVENTS
Provincial Department of Water Resources and Meteorology, with the Provincial Department of Agriculture (PDAFF)	15 November 2026	staff time	a failed March crop, a second round of replanting debt, and drinking water lost to rice

In 2025 Battambang planned 27,000 ha of dry-season rice and farmers planted about 86,115 ha. In January 2024 the governor halted the second and third rounds to keep water for daily use and livestock. In 2026 drought hit 18,362 ha by 1 July, mostly in Sangkae and Ek Phnom, and then the Sek Sak reservoir overflowed into the Sangkae within two weeks: the same reservoirs face drought and flood in one season, so one operating rule should cover both.

Transferability: administrative precedent in Battambang and Pursat. Enforcement is weak; farmers ignored advisories in 2019 and 2024.

7. Prepare the referral hospital and health centres for heat stroke by March, and review medicines for older people with high blood pressure or diabetes in February

PARTIALLY CONFIRMED | PUBLIC HEALTH · PHILANTHROPIST | ADAPTATION INVENTORY: NOT CHECKED

OWNER	BY WHEN	COST	PREVENTS
Provincial Health Department, with Battambang Provincial Referral Hospital and the health centres	medicine reviews February 2027; kit in place by 1 March 2027	KHR 163.2 million (US$40,000), released against an itemised request (placement 3)	deaths from heat stroke that a cooled bed and fluids would have prevented

* Suggested formulation: check with a health worker before sharing.

About 17–20% of Cambodian adults have raised blood pressure and 6.3–6.4% raised glucose or diabetes (STEPS 2023). Health centres mostly screen and refer, and drug stock-outs are documented. A kit means ice, intravenous fluids, a cooled triage area, and ORS at health centres before April.*

Transferability: reasoned from how heat kills, not from a trial. The cost has no published basis and could duplicate a ministry budget line, so it waits for an itemised request.

8. Start counting now: collect weekly deaths by age from commune registers, so the March–May toll can be measured against normal

PARTIALLY CONFIRMED | PUBLIC HEALTH | ADAPTATION INVENTORY: NOT CHECKED

OWNER	BY WHEN	COST	PREVENTS
Provincial Health Department, with commune and sangkat civil registration offices	from 1 December 2026	staff time	a heat season whose deaths are never seen, and actions nobody can evaluate

Cambodia's certified heat deaths are near zero: in April 2024, near 42°C, Phnom Penh's health department reported none. About 70% of deaths are registered (ESCAP 2025, with caution), and Cambodia publishes no vital statistics reports, so weekly counts by age would have to be requested from the Ministry of Interior's identification department and the commune offices.

Transferability: standard excess-mortality practice. Feasible only as a local data request; it is not an existing feed.

9. Stock guppies in water jars from February to April and watch hospital dengue counts from January; add lightning safety to the same messages for the first storms

PARTIALLY CONFIRMED | PUBLIC HEALTH · HISTORIAN · SKEPTIC · EMERGENCY MANAGER · PHILANTHROPIST | ADAPTATION INVENTORY: NOT CHECKED

OWNER	BY WHEN	COST	PREVENTS
Provincial Health Department, with commune councils and VHSG volunteers	hospital watch from January; jar campaign February – April 2027	not costed here; belongs in the rainy-season dengue budget	a dengue surge when the rains return, and lightning deaths in open fields

* Suggested formulation: check with a health worker before sharing.

Guppies plus community engagement roughly halved adult female dengue mosquitoes in a Kampong Cham trial (abundance ratio 0.49, 95% CI 0.31–0.77); adding pyriproxyfen did no better (Hustedt 2021). Jars hold water through the dry season, so stocking them before the rains cuts the mosquito population. In January 1998, after the 1997 El Niño, dengue cases were almost five times the epidemic threshold, months before the 1998 epidemic of about 16,200 cases and 475 deaths. Cambodia recorded 63,106 cases in 2025 and was 58% higher again by July 2026.

Transferability: Cambodian trials, but they measured mosquitoes, not cases. Dengue peaks from May to October, after the heat, so this is preparation for the rains rather than a heat measure.

10. Ask ARDB to require a published floor price from Battambang mills that draw on the KHR 816 billion (US$200 million) paddy credit, for late-planted paddy bought in December and January

PARTIALLY CONFIRMED | FOOD SUPPLY CHAIN | ADAPTATION INVENTORY: NOT CHECKED

OWNER	BY WHEN	COST	PREVENTS
Agricultural and Rural Development Bank (ARDB), with the rice mills and PDAFF	**30 November 2026**	**no public cost; a condition on existing credit**	**distress selling at harvest that pushes farmers into debt**

On 12 September 2026 the government announced up to KHR 816 billion (US$200 million) in ARDB working-capital loans for mills to buy and store paddy, at a meeting with Battambang, Pursat, and Banteay Meanchey millers. Farmers without drying or storage "have to sell immediately". The package sets no floor price, and economists warned the benefit may not reach farmers without monitoring.

Transferability: a condition we propose, not an existing rule. Rice supply is ample, so this protects income, not food supply.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	Provincial Health Dept	Referral hospital & health centres	Village chiefs & VHSG	HelpAge OPAs	Commune & sangkat councils	Provincial Education Dept	NBC	Lenders (CMA & ABC)	Ministry of Planning & MoSVY	NCDM & WFP	PCDM	PDWRAM	PDAFF	ARDB & rice mills
1. Daily checks on people at risk	**A**	**I**	**R**	**R**	**C**									
2. School heat schedule, pre-signed	**C**					**A/R**					**C**			
3. Repayment pause, Mar–May					**I**		**A**	**R**		**C**				
4. IDPoor catch-up and cash trigger			**C**		**R**				**A**	**R**				
5. Water for communes off the network		**I**			**R**						**A**	**R**		
6. Cap dry-season rice by water					**C**						**C**	**A**	**R**	
7. Heat-stroke readiness	**A**	**R**	**C**											
8. Weekly death count	**A**				**R**									
9. Dengue jars and storm safety	**A**	**R**	**R**		**R**									
10. Floor price on mill credit													**C**	**A/R**

What the chart shows: the Provincial Health Department is accountable for four of the ten rows and carries the heat response. Rows 3, 4, and 10 need decisions in Phnom Penh, by the central bank, the planning and social ministries, and ARDB, rather than money in Battambang; village chiefs and VHSG volunteers carry most of the door-to-door work.

WHERE MONEY DOES THE MOST

FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per dollar. The envelope is bounded by what this place can spend before the peak, not by what it needs.

FLOOR

KHR 102.0 million (US$25,000)

TARGET

KHR 387.7 million (US$95,000)

CEILING

KHR 550.9 million (US$135,000)

LARGEST UNCERTAINTY

whether HelpAge's older people's associations are active in the city's sangkats

Floor = daily-check pilot for about 5,000 people at KHR 20,400 (US$5) each, where HelpAge confirms active associations (KHR 102.0 million (US$25,000); the unit cost is an estimate). Target = floor + scale-up to about 11,000 people over 60 in the city, conditional on documented coverage (KHR 122.4 million (US$30,000); the 2019 census counted 119,251 people in the municipality, and 9–10% are over 60) + heat-stroke

* Suggested formulation: check with a health worker before sharing.

readiness at the referral hospital and health centres, released against an itemised request (KHR 163.2 million (US$40,000)) = KHR 387.7 million (US$95,000). Ceiling = target + checks on about 5,000 older people in the peri-urban communes (KHR 102.0 million (US$25,000)) + a water-trucking contingency for communes outside the piped network (KHR 61.2 million (US$15,000)) = KHR 550.9 million (US$135,000). The target fell from KHR 612.1 million (US$150,000) in verification (−37%), because ceramic filters duplicated the new piped network and dengue control belongs in the rainy-season budget.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
HelpAge Cambodia, through older people's associations with confirmed coverage	Action 1: daily-check pilot, about 5,000 people	KHR 102.0 million (US$25,000)	direct grant	KHR 20,400 (US$5) per person, an estimate; Ahmedabad heat action plan averted about 1,190 deaths a year (Hess 2018), for a whole-city plan	Partially confirmed
HelpAge Cambodia, if city coverage is documented	Action 1: scale-up to about 11,000 people over 60 in the city	KHR 122.4 million (US$30,000)	top-up, conditional	Same unit cost; 2019 census, 119,251 people in the municipality	Partially confirmed
Provincial Health Department, for Battambang Provincial Referral Hospital and health centres	Action 7: heat-stroke kit	KHR 163.2 million (US$40,000)	restricted grant against an itemised request	No costed study; amount is the lens's judgement	Unverified
HelpAge Cambodia, older people's associations in peri-urban communes	Action 1: checks on about 5,000 more older people	KHR 102.0 million (US$25,000)	direct grant	Same unit cost, an estimate	Partially confirmed
PCDM, for communes outside the piped network	Action 5: water-trucking contingency	KHR 61.2 million (US$15,000)	contingency, released if ponds run dry	Lens estimate; no published trucking cost for Battambang	Unverified

* Suggested formulation: check with a health worker before sharing.

The funder's one thing: the big money here is already committed, from the KHR 486 billion (US$119 million) water project to the Family Package. The gap is small and human: a few volunteers per village who know which grandmother lives alone, signed up before the heat. The most valuable decisions in this briefing, time on loans and a pre-signed school order, cost no money at all.

Do not do this

CONTRAINDICATED · ADVICE THAT WOULD COST LIVES HERE

Advice and practice that circulate for drought, heat, and dengue in Cambodia and would cost lives, land, or money here. Each was checked by a verifier.

Do not take a new loan to replant a lost crop

CONTRAINDICATED VERIFIER: CONFIRMED

Most microloans are secured on land titles, the central bank's loan relief has ended, and remittances from Thailand fell by about a quarter. Replanting after the July 2026 drought "doubles costs while the debt remains" (Cambodian farmer federation). In March 2026, 85% of indebted returnees surveyed could not repay.

Instead: Ask the lender to restructure before a missed payment (action 3, and the For you section).

Do not pay for pumps or fuel to grow a second or third dry-season crop

CONTRAINDICATED VERIFIER: PARTIALLY CONFIRMED

In 2025 farmers planted three times the dry-season area the province planned, and paddy far from water was lost even after pumping. In January 2024 the governor had to suspend the later rounds to keep water for drinking and livestock. Pumping draws down the water households need in April.

* Suggested formulation: check with a health worker before sharing.

Instead: Allocate canal water by command area, publish the limits (action 6), and advise low-water crops. Paying farmers to skip a round has no Cambodian precedent and is untested.

Do not rely on fogging or Abate (temephos) alone against dengue

CONTRAINDICATED VERIFIER: PARTIALLY CONFIRMED

Dengue mosquitoes from eight Cambodian sites resisted temephos, and seven of eight resisted deltamethrin; permethrin barely killed them (Boyer 2018). Temephos has been the main larvicide since 1992 and is still promoted, so it is not banned, but on its own it is unreliable.

Instead: Guppies in large jars and lids on small ones, with community engagement (action 9).

Do not drink from a new or deepened tube well before it has been tested *

CONTRAINDICATED VERIFIER: PARTIALLY CONFIRMED

Arsenic in the Tonle Sap provinces is usually low but patchy, with local hot spots up to 100 µg/L, ten times the WHO limit, and a WHO-sponsored survey in 2000 found "relatively high" levels in some Battambang samples. Drilling is a common drought response.

Instead: Test any new well for arsenic, fluoride, salinity, and E. coli before use, and prefer rehabilitating ponds and connecting to the piped network.

Do not tell older people to stop using fans in the heat *

CONTRAINDICATED VERIFIER: CONFIRMED

Some US and UK guidance discourages fans above 35°C. Modelling suggests fans still help older people up to about 38°C, or 37°C for people on anticholinergic drugs, and harm mainly in very hot, dry air (Morris 2021). Battambang's April highs average 35–37°C. The 2024 energy-saving advice asked people to use less air conditioning, not to switch off fans.

* Suggested formulation: check with a health worker before sharing.

Instead: A fan with wet skin; above about 38°C, move older people to a cooler place (For you section).

Do not judge the heat response by the official count of heat deaths

CONTRAINDICATED VERIFIER: CONFIRMED

In April 2024, near 42°C, Phnom Penh's health department reported zero heat deaths. About 30% of deaths in Cambodia go unregistered, and heat is rarely written on a death certificate. A zero count will be read as success.

Instead: Measure excess deaths from all causes against a weekly baseline (action 8).

WHAT HAS HAPPENED HERE BEFORE — THE RECORD

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
1997–98	El Niño, then dengue: in January 1998 cases were almost five times the epidemic threshold (WHO)	about 16,200 cases and 475 deaths nationally in 1998 (US embassy cable; sources vary slightly)	National dengue programme built on temephos, now met with resistance
2013	Floods; Battambang among the worst-damaged provinces (NCDM)	168 or more dead nationally	Nothing specific to Battambang found
2015–16	El Niño drought, the worst since 1979; 18 of 25 provinces declared water shortages	2.5 million people affected; more than 49,000 ha of Battambang crops damaged	Water trucked after shortages were declared, not before
2019	Drought dries Sangkae's paddies; record dengue year	68,597 dengue cases and 48 deaths nationally	Reservoir water withheld for drinking
2020	October floods; Battambang the hardest-hit province	31–44 dead nationally; 13,180 Battambang households affected, 4,592 families displaced	Flood cash transfers through IDPoor

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
2024	January: governor halts the second and third dry-season rice rounds. April: about 43°C nationally, the hottest in about 170 years; school hours cut	zero heat deaths reported in Phnom Penh; 68 lightning deaths nationally that year	One-off school order, not a standing rule
2025	Dry-season rice planted on 86,115 ha against 27,000 planned; Thai border closed in June; hundreds of thousands of workers return through Battambang	63,106 dengue cases and 79 deaths nationally	Relief directive for returnees, extended to 31 March 2026
2026	Drought on 18,362 ha by 1 July, then the Sek Sak reservoir overflows two weeks later; September floods; household water put first on 19 October	Nationally, 302,121 ha of rice affected by drought and 92,135 ha by floods by October	Central bank declines to extend loan relief (September)

The recurrence nobody plans for: Battambang's disasters come in pairs. Drought and flood struck within two weeks in July 2026, through the same reservoirs, and every plan treats them as separate emergencies. The second pattern is quieter: each dry year since 2016 has been carried on credit, and the credit is secured on land.

WHERE THE EXPERTS DISAGREE

UNRESOLVED

- **Older people have been left alone by migration** versus **migration, not desertion**. The frontline lens leads with grandparents raising children alone; the public health lens cites a Battambang survey that found migration had only modest effects on older people's care, and the 2025 border closure brought many adults home. The list in action 1 should be built door to door rather than assumed.
- **Pay a cash top-up** versus **give time on loans instead**. The food lens would top up the Family Package; the economist argues that a repayment pause is cheaper and protects the land title, which is the household's real insurance. The report keeps both: the pause as action 3, the cash trigger as an ask in action 4.
- **Pagodas as cool refuges** versus **no evidence that anyone would go**. Visiting cool places was protective in heat-wave studies (odds ratio 0.34), but no Cambodian evidence on pagodas or monks as heat outreach was found. Pilot it within action 1 rather than build on it.

* Suggested formulation: check with a health worker before sharing.

WHAT NOBODY COVERED

BLIND SPOT

Two gaps. First, the forecast ends in March, and April and May are hotter; no lens could put a number on those months, so the real toll this season is likely higher than the figure shown. Second, no lens looked at people working in heat for an employer: construction workers, brick-kiln workers, and market porters. Cambodia's labour ministry heat regulation is still a draft, and no binding rule on working hours in heat exists.

HOW TO READ THIS

METHOD

The panel is author-built. Eight simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result.

Mortality figures are resolved at province level here, Battambang province, not the city. The El Niño projection covers *heat only* and stops in March; drought, crop loss, dengue, and lightning are additional to it and are carried here by the historical record rather than by a published number.

HOW THIS BRIEFING WAS PRODUCED

METHODOLOGY

- **Scope.** Chosen on El Niño Ready's initiative, not from a request: Battambang has the highest forecast heat burden of any Cambodian province on the Climate Impact Lab tracker. The briefing covers Battambang municipality (Krong Battambang, 119,251 people in 2019) and the nearby rice communes of Sangkae and Ek Phnom. The tracker resolves Cambodia at province level, so no city-only death figure exists.
- **Exposure.** Climate Impact Lab El Niño Heat Mortality Tracker data, file `v261026_adm1_output_BASE_rounded_cust.csv` (version of 26 October 2026, pre-peer-review): Batdâmbâng, 150 additional heat deaths June–September 2026, 400 forecast October 2026 – March 2027 (October 21, November 31, December 27, January 42, February 98, March 180), 550 in total; Cambodia, 6,300; the world, 857,400. This update supersedes the September report *451,000 at Risk*. Method: ECMWF SEAS5 seasonal forecasts applied to temperature–mortality relationships against a 1996–2025 baseline. The figures count heat only; we inferred this from the data, because cold countries show exactly zero in every winter month. They are never added to the Impact Map's long-run projections.
- **Expert panel.** Eight simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, philanthropist, and food supply chain analyst) each researched this place independently and in parallel, from web sources, to a fixed brief.

* Suggested formulation: check with a health worker before sharing.

- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias before it counted in an action's favour; the heat evidence, for example, comes from France, the United States, and India, and is labelled as such. Contradictions were kept, not resolved by dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, each was given a named payer and trigger or marked unfunded, and the funding placements were reconciled against the ranked list.
- **Verification.** Six verifier agents checked every recommendation on three tests: evidence, transferability, and feasibility here before the peak. Financial actions were also checked for availability in Cambodia, a named payer, a trigger, and whether they already exist; food actions for live price series and timing; the funding envelope for real recipients, published unit costs, and duplication, then re-summed.
- **Who does what.** The RACI chart is built from the owners named in the verified actions, with one accountable party per action. It is a proposal: no organisation named in it has been contacted or has agreed to a role.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Not consulted in this run. The inventory loads its data in the browser, and every verifier saw an empty list. No action here should be read as having no precedent; none was checked.
- **Money.** Costs are shown in Cambodian riel (KHR) with US dollars in parentheses, at KHR 4,081 = US$1, the National Bank of Cambodia official rate for 30 October 2026 (as published by the General Department of Taxation). US dollars also circulate widely in Cambodia, and several sources quote dollars only.
- **Reader.** No requester; the briefing was built on El Niño Ready's initiative, so the For you section is written for a resident. Its steps were taken from the verified actions and from what the panel found about households here, and each was verified on the same three tests. The rest of the report is written for the province, the city, and their partners.
- **Health wording.** Passages where a wording error could hurt someone (drinking-water treatment and well water, medicines, cooling and fans, and signs of heat stroke) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.
- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so. Where a cost is an estimate, it says so. Every source is listed below.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. No Battambang-only death toll exists for any heat wave, flood, or dengue year; most figures are national. The forecast stops in

* Suggested formulation: check with a health worker before sharing.

March. HelpAge's coverage of the city, the current Family Package amounts, Battambang's IDPoor household count, and the Kamping Puoy reservoir's history could not be confirmed. One link (CCAFS) failed a security check and was not opened. The locator map shows Battambang at country scale.

References · Exposure: Climate Impact Lab El Niño Heat Mortality Tracker data v261026 (downloaded 30 Oct 2026); 451,000 at Risk (superseded) · confirmed
Heat and health: Bouchama 2007; Hess 2018; Morris 2021; MoH heat advice 2023; April 2024 heat and schools; school hours; 2024 energy advice; STEPS 2023; ESCAP CRVS; HelpAge OPAs; CDRI migration study · partially confirmed
Water and drought: household water priority, 19 Oct 2026; NCDM July 2026; Battambang drought 2026; Sek Sak overflow; NCDM 9-month totals; WFP 2016; Battambang water project; 2025 dry-season rice; 2024 suspension; arsenic map; EWS 1294 · partially confirmed
Debt, cash, and food: NBC forbearance refused; AMRO NPLs; returnee relief; CENTRAL 2026; remittances; HRW 2025; On-Demand IDPoor; Family Package; UNDP at-risk cash; PRISM; ARDB mill credit; FAO GIEWS; exchange rate · partially confirmed
Dengue and storms: Hustedt 2021; Seng 2008; Boyer 2018; WHO 1998; dengue 2026; lightning 2024; 2020 floods · partially confirmed
Exposure data · Climate Impact Lab · Impact Map · Adaptation Inventory
Generated 30 October 2026 for Battambang City, Battambang Province. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

* Suggested formulation: check with a health worker before sharing.